

Algorithms

Algorithm to find the 2's complement of a number

Suppose we want to find the 2's complement of number: **01101100**

We assume:

0	1	1	0	1	1	0	0	<- number
7	6	5	4	3	2	1	0	<- bit position

MAX: holds the position of the most-significant bit in the number

Posi: holds the position of the bit being processed

FlipIt: holds **true** if the bits need to be flipped
holds **false** if the bits need not to be flipped

procedure 2sComplement

set MAX to 7

set FlipIt to false

set Posi to 0

while (Posi <= MAX) do

 if (bit @ Posi = 0 AND FlipIt = false) then

 keep the zero (0)

 else if (bit @ Posi = 1 AND FlipIt = false) then

 keep the one (1)

 set FlipIt to true

 else

 flip bit @ Posi

 increment Posi

procedure 2sComplement

set MAX to 7

set FlipIt to false

set Posi to 0

while (Posi <= MAX) do

 if (FlipIt = false) then

 if (bit @ Posi = 1) then

 set FlipIt to true

 keep bit as it is

 else

 flip bit @ Posi

 increment Posi

; most-significant bit is at position 7

; do not start flipping bits yet

; start processing at least-significant bit

; while we have not processed all bits

; if we don't need to flip bits yet

; if bit at current position is a one (1)

; start flipping bits from now on

; do nothing

; otherwise

; flip the bit at current position

; move on to the next bit